

HARINI



HIGHSCORE 2500



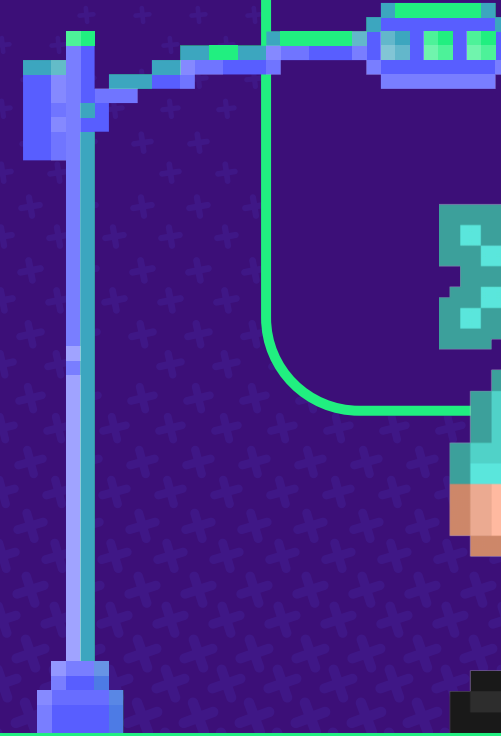
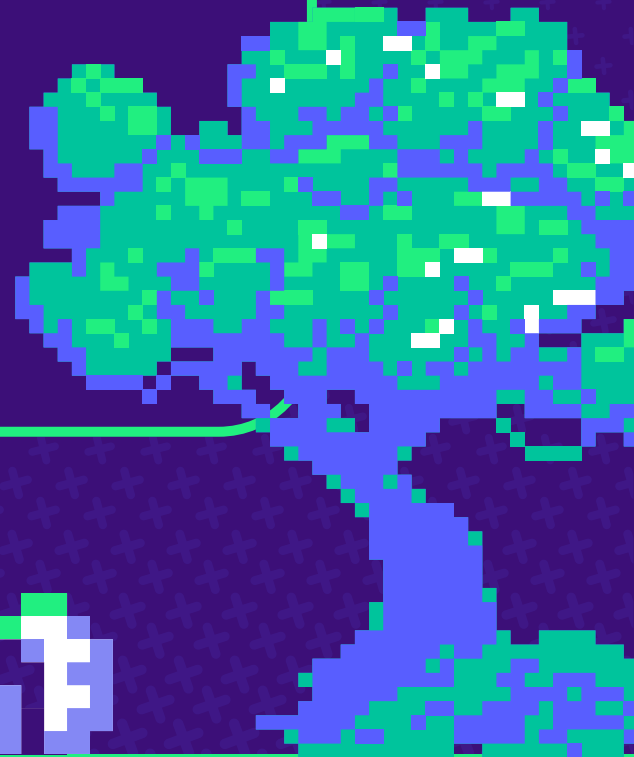
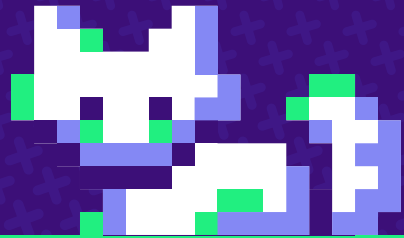
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DANCEPENA

START

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SIGN IN



MENU



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INSPIRATION

✦ WE DREW INSPIRATION FROM ARCADE DANCE RHYTHM GAMES AND WHAC-A-MOLE, BOTH OF WHICH RELY ON QUICK REACTION, TIMING, AND PHYSICAL INTERACTION. FROM THE DANCE ARCADE GAME, WE ADOPTED THE IDEA OF FOLLOWING ILLUMINATED DIRECTIONS IN SYNC WITH MUSIC. FROM WHAC-A-MOLE, WE BORROWED THE CONCEPT OF REACTING INSTANTLY TO RANDOMLY APPEARING CUES. WE THEN TRANSLATED THESE LARGE, PHYSICAL ARCADE EXPERIENCES INTO A COMPACT, SENSOR-BASED VERSION USING AN ESP32. BY REPLACING FULL-SCALE FLOOR PANELS WITH DIRECTIONAL TOUCH INPUTS AND LED INDICATORS, WE RETAINED THE ENERGY AND RESPONSIVENESS OF ARCADE GAMEPLAY WHILE MAKING IT MINIMAL, PORTABLE, AND TECHNICALLY ACHIEVABLE WITHIN OUR PROJECT CONSTRAINTS.



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GAME FLOW

➤ SYSTEM START

- THE DEVICE POWERS ON AND WAITS FOR THE PLAYER TO SELECT A DIRECTION

➤ SONG SELECTION

- THE PLAYER TAPS ONE OF THE FOUR DIRECTIONS (UP, DOWN, LEFT, RIGHT).
- EACH DIRECTION UNLOCKS A DIFFERENT BEAT PATTERN.

➤ GAME ACTIVATION

- THE SELECTED RHYTHM STYLE IS ACTIVATED.

➤ RANDOM STEP GENERATION

- THE SYSTEM GENERATES A RANDOM DIRECTION FOR THE PLAYER TO FOLLOW.

➤ VISUAL CUE

- THE CORRESPONDING LED LIGHTS UP, INDICATING WHICH DIRECTION TO PRESS.

➤ USER RESPONSE CHECK

- THE SYSTEM WAITS FOR THE PLAYER TO PRESS A DIRECTION.

➤ CORRECT INPUT → BEAT PROGRESSES

➤ INCORRECT INPUT → GAME OVER

➤ CONTINUOUS LOOP

- THE CYCLE CONTINUES INDEFINITELY AS LONG AS THE PLAYER KEEPS PRESSING THE CORRECT DIRECTIONS.

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PARTS AND WHY?

➦ TOUCH SENSORS CAPTURE USER INPUT.

➦ LEDS PROVIDE VISUAL GUIDANCE.

➦ THE BUZZER GENERATES RHYTHMIC PATTERNS USING DIFFERENT FREQUENCIES.

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WHAT LOGICS AND WHY?

➡ WHILE LOOPS TO CONTROL CONTINUOUS
GAMEPLAY

➡ IF-ELIF CONDITIONS TO DETECT INPUT

➡ RANDOM NUMBER GENERATION TO CREATE
STEP PATTERNS

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PAIN POINTS

➦ BUZZER VOLUME WAS TOO LOW FOR A PROPER ARCADE EXPERIENCE.

➦ CIRCUIT WIRING BECAME MESSY DUE TO TOO MANY CONNECTIONS.

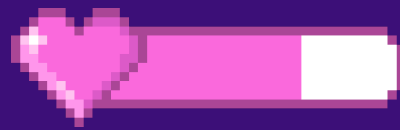
➦ DIFFICULTY IN TRACKING AND ASSIGNING CORRECT GPIO PINS.

➦ SENSORS SOMETIMES GAVE INCONSISTENT OR FALSE TRIGGERS.

➦ SYNCING LED LIGHTS, BEATS, AND INPUT TIMING WAS CHALLENGING.

➦ LIMITED SOUND QUALITY DUE TO BASIC BUZZER HARDWARE.

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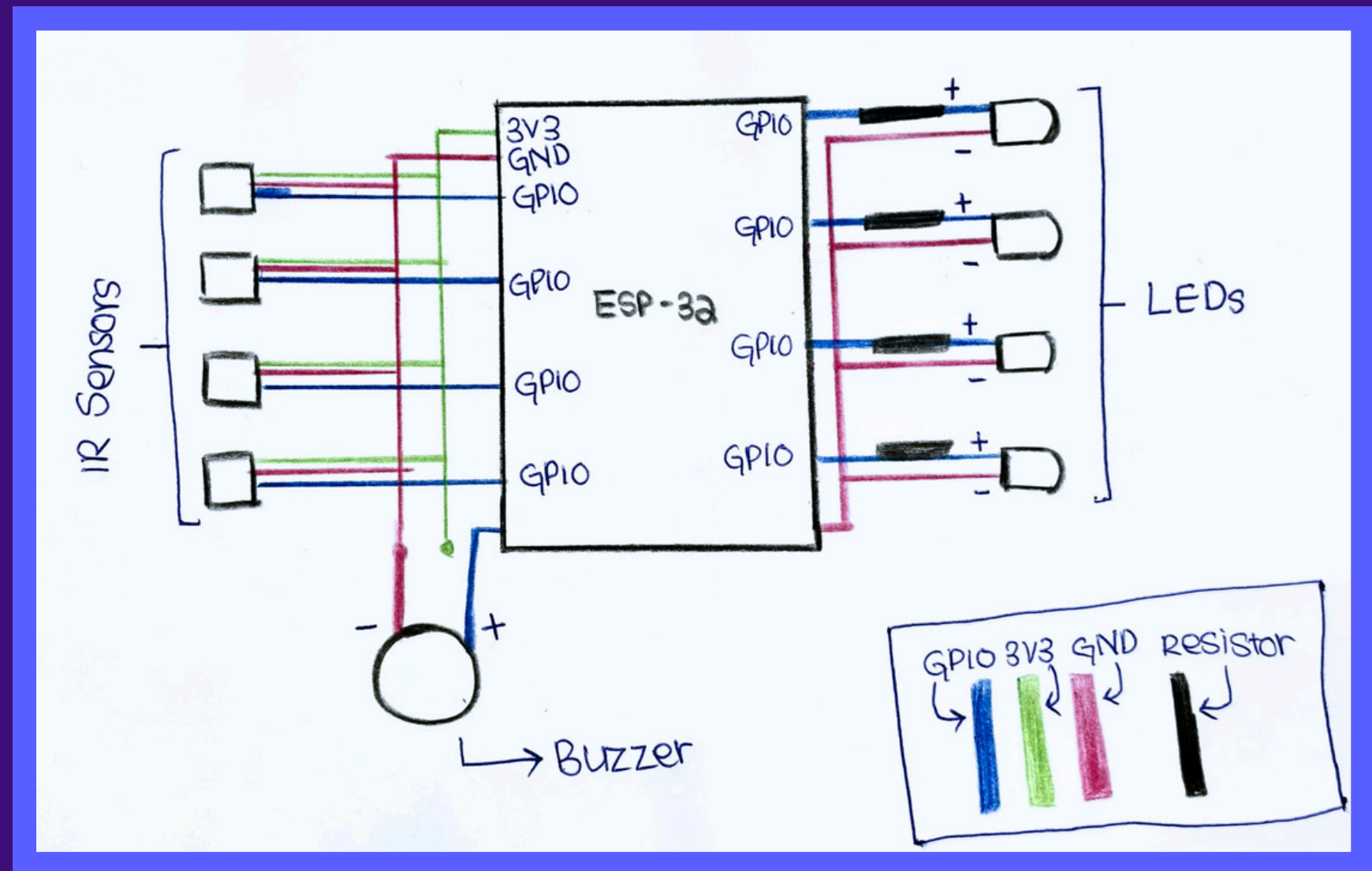
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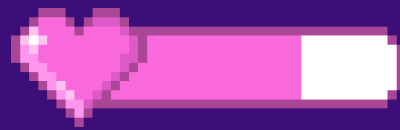
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CIRCUIT DIAGRAM



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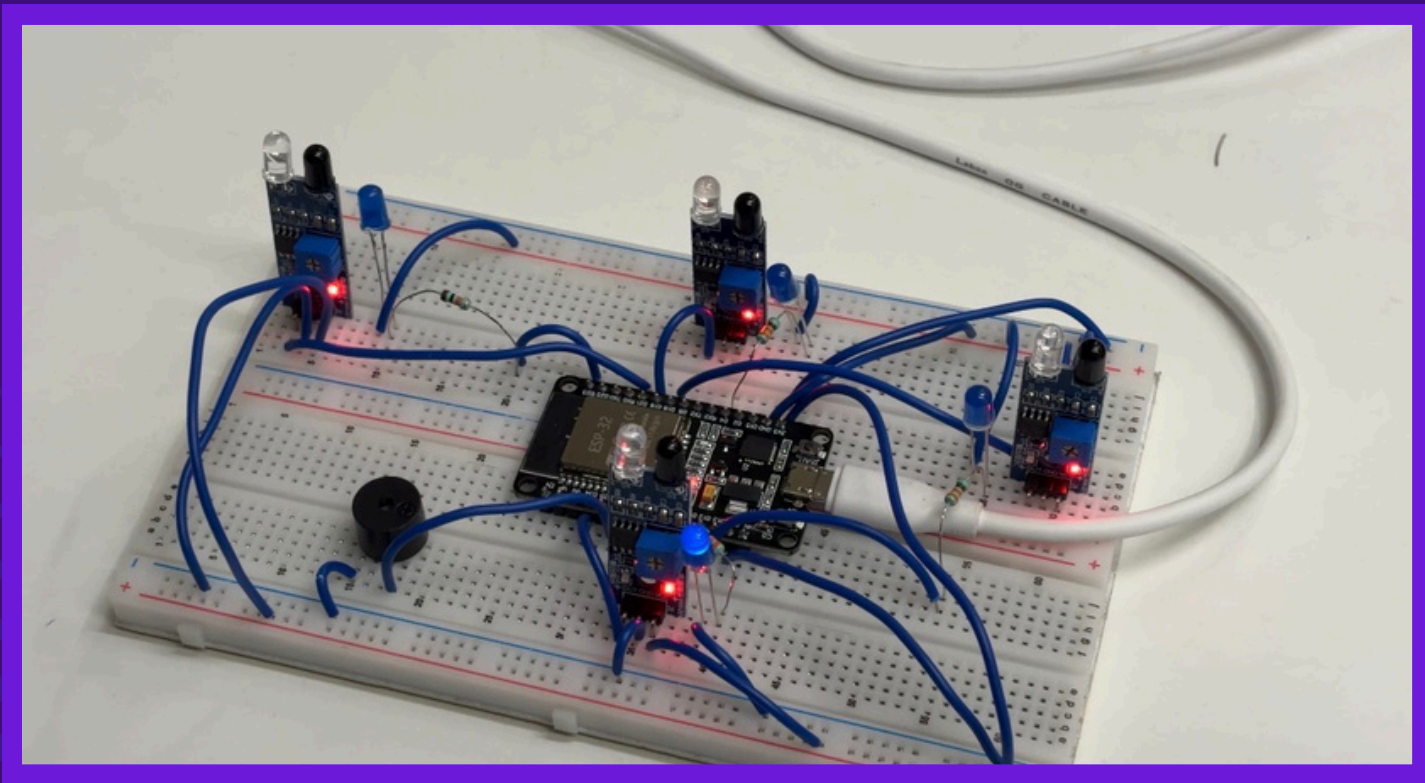
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PICTURES AND VIDEOS



INDIVIDUAL CONTRIBUTION

```
#harini
buzzer = PWM(Pin(32))
buzzer.duty_u16(2000)
```

C5 = 523

D5 = 587

E5 = 659

G4 = 392

A4 = 440

G5 = 784

```
#harini
buzzer.duty_u16(2000)
if song == 1:
    buzzer.freq(C5)
    time.sleep(0.15)
    buzzer.freq(E5)
    time.sleep(0.15)
    buzzer.freq(G5)
    time.sleep(0.25)
elif song == 2:
    buzzer.freq(G4)
    time.sleep(0.25)
    buzzer.freq(A4)
    time.sleep(0.15)
    buzzer.freq(G4)
    time.sleep(0.25)
elif song == 3:
    buzzer.freq(C5)
    time.sleep(0.1)
    buzzer.freq(C5)
    time.sleep(0.1)
    buzzer.freq(D5)
    time.sleep(0.1)
    buzzer.freq(E5)
    time.sleep(0.2)
elif song == 4:
    buzzer.freq(E5)
    time.sleep(0.25)
    buzzer.freq(D5)
    time.sleep(0.2)
    buzzer.freq(C5)
    time.sleep(0.3)
buzzer.duty_u16(0)

led_up.value(step == 1)
led_down.value(step == 2)
led_left.value(step == 3)
led_right.value(step == 4)
time.sleep(0.4)
led_up.value(0)
led_down.value(0)
led_left.value(0)
led_right.value(0)
```

```
#harini

elif left.value() == 0:
    if step == 3:
        pressed = 1
    else:
        wrong = 1
        pressed = 1
elif right.value() == 0:
    if step == 4:
        pressed = 1
    else:
        wrong = 1
        pressed = 1
    time.sleep(0.01)
    i = i + 1

if wrong == 1:
    buzzer.freq(200)
    buzzer.duty_u16(4000)
    time.sleep(0.5)
    buzzer.duty_u16(0)
```

- CIRCUIT BUILDING
- PPT

INDIVIDUAL CONTRIBUTION

```
#stuti
from machine import Pin, PWM
import time
import random

up = Pin(19, Pin.IN, Pin.PULL_UP)
down = Pin(34, Pin.IN, Pin.PULL_UP)
left = Pin(21, Pin.IN, Pin.PULL_UP)
right = Pin(18, Pin.IN, Pin.PULL_UP)

led_up = Pin(4, Pin.OUT)
led_down = Pin(14, Pin.OUT)
led_left = Pin(23, Pin.OUT)
led_right = Pin(13, Pin.OUT)
```

```
#stuti
while True:

    song = 0
    while song == 0:
        if up.value() == 0:
            song = 1
        elif down.value() == 0:
            song = 2
        elif right.value() == 0:
            song = 3
        elif left.value() == 0:
            song = 4
        time.sleep(0.05)
    time.sleep(0.5)

    wrong = 0
    while wrong == 0:
        pattern = [random.randint(1, 4) for i in range(5)]
        i = 0
        while i < 5 and wrong == 0:
            step = pattern[i]
```

- CIRCUIT BUILDING
- EXTERIOR AESTHETICS

```
pressed = 0
while pressed == 0:
    if up.value() == 0:
        if step == 1:
            pressed = 1
        else:
            wrong = 1
            pressed = 1
    elif down.value() == 0:
        if step == 2:
            pressed = 1
        else:
            wrong = 1
            pressed = 1
```

MENU



THANK YOU!

